

Excerpt from “Sensing with Earables: A Systematic Literature Review and Taxonomy of Phenomena”

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ABSTRACT

By adding sensing capabilities to ear-worn devices, earables have emerged as a new platform. The ears are located closely to a number of important anatomical structures (e.g., brain, blood vessels). Also, the ear canal deforms upon facial movements and the ears can be comfortably touched by the hands. In a recent paper, we conducted a systematic literature review of 271 earable papers. We synthesized an open-ended taxonomy of 47 phenomena that can be sensed in, on, or around the ear. We identified 13 fundamental phenomena from which all other phenomena can be derived, and discuss sensing principles to detect them. The phenomena were reviewed in-depth in four main areas: (i) physiological monitoring and health, (ii) movement and activity, (iii) interaction, and (iv) authentication and identification. This breadth highlights the potential earables have to offer as a ubiquitous, general-purpose platform.

CCS CONCEPTS

• General and reference → Surveys and overviews.

KEYWORDS

earables, hearables; earphones; headphones; earbuds; ear wearable; earpiece; ear-worn; ear-mounted; ear-attached; ear-based

ACM Reference Format:

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1 INTRODUCTION

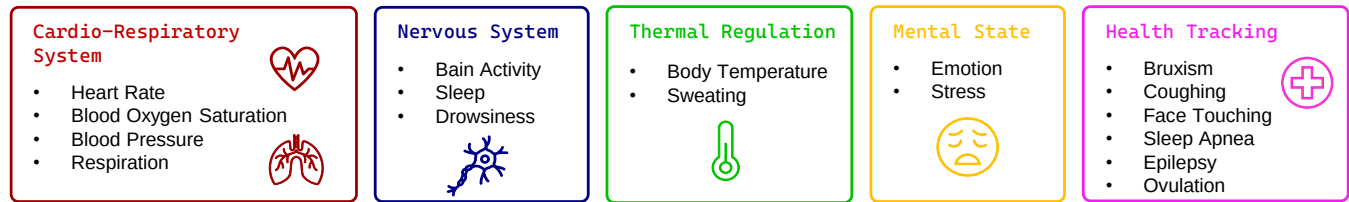
Earables are wearable that are worn in or near the ears. Hearing aids and personal speakers are examples of ear-worn electronics that have emerged for specific uses. As a new class of ubiquitous computing platform, we define “earables” as devices that integrate broader features. Due to their distinctive placement on the human body, earables (also known as “hearables”) offer a unique platform for sensing a variety of properties, processes, and activities. At the heart of much of the research in this new field are questions of sensing - what can be detected and observed with earables, and what interactions and applications are enabled by sensing in or on the ear?

In a recent paper, we conducted a systematic and comprehensive literature review of 271 earable publications, representing the state of the art in earable sensing. This excerpt gives a brief introduction to article No. 135 published in Volume 6, Issue 3 in the *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* [1].

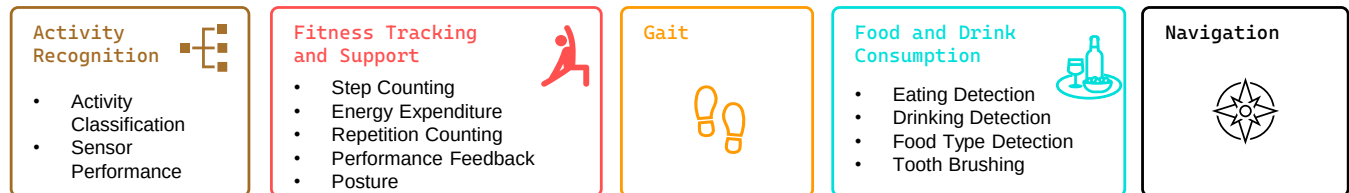
2 METHODOLOGY

Inspired by related reviews, we followed a structured process by collecting and filtering papers from the ACM and IEEE digital libraries using a number of defined selection criteria (e.g., sensing has to occur on or around the ears). This was supplemented with backward chaining using the same criteria applied to papers from other digital libraries (e.g., Springer). We used the following guiding definition of earables: “Earables are devices that attach in, on, or in the immediate vicinity of the ear to offer functionalities beyond basic audio in- and output.” The described process resulted in 271 relevant articles which we analysed and clustered, and from which we derived a novel earable taxonomy consisting of different phenomena.

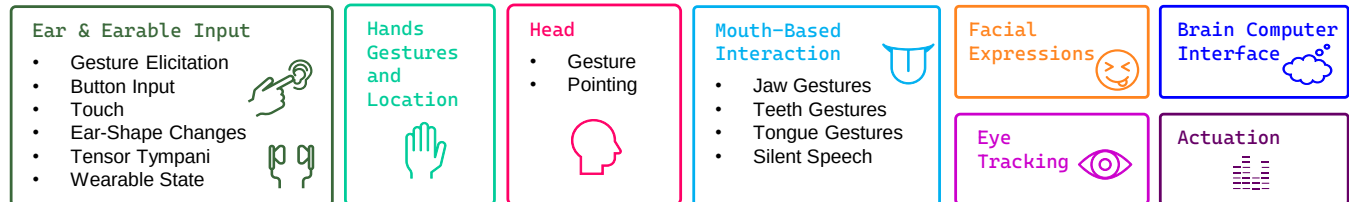
PHYSIOLOGICAL PARAMETERS AND HEALTH



MOVEMENT AND ACTIVITY



INTERACTION



AUTHENTICATION AND IDENTIFICATION



Figure 1: Overview map of earable sensing. The map is organised by phenomena that can be captured with earable sensors.

3 CONTRIBUTION

Each paper in the dataset was categorized with respect to the applied sensing principle, the corresponding phenomena which was sensed, and the application for which it was used. Through multiple iterations, we then synthesized a higher-level taxonomy of phenomena sensed with earables that is open-ended and extensible (new sensors might emerge, and further phenomena explored).

At the lowest level, we then identified and characterised phenomena that are directly sensed with sensors placed in or on the ear as *fundamental phenomena*, including for instance motion, body temperature and blood perfusion. Other phenomena are identified as indirectly observable and derived from fundamental phenomena, ranging from physiological parameters (e.g., heart rate) and lower-level cues (e.g., earable state; in or out of ear) to conditions (e.g., stress), actions (e.g. gestures), activities (e.g. daily tasks) and other context (e.g. user identity). In total, we identified and categorised close to 50 phenomena. We showed how higher-level phenomena build on 13 fundamental phenomena, and relate this to 21 different sensors and sensing principles that have been investigated for their observation.

Figure 1 provides a simplified map of these phenomena, clustered into related themes and grouped into four main areas of research: (1) physiological parameters and health; (2) movement and activity; (3) interaction; and (4) authentication and identification. For each of these research areas, we provide an extensive overview of the research conducted in the space.

Further sensing developments will likely expand the already rich set of phenomena sensed by earables, and new applications will emerge that leverage those currently available. However, to-date most earable sensing research has not been rigorously tested in-the-wild. Future work will have to demonstrate ecological validity and overcome robustness and engineering challenges to unleash the full potential that earables have to offer as a ubiquitous, general-purpose platform.

REFERENCES

- [1] Tobias Röddiger, Christopher Clarke, Paula Breitling, Tim Schneegans, Haibin Zhao, Hans Gellersen, and Michael Beigl. 2022. Sensing with Earables: A Systematic Literature Review and Taxonomy of Phenomena. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 3 (Sep 2022), 135.